25901126 MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION March 2026

Fourth SEMESTER Discipline Specific Elective (DSE) - MG4DSEBOTA05

BIO-INSTRUMENTATION AND QUALITY ANALYSIS 2

PART A One or Two Sentences

Answer any 10 questions. Each question carries 2 marks.

1. What are the commonly used detergents for cell disruption? [K] / [CO1]

Ionic detergents like **SDS** and non-ionic detergents like **Triton X-100**, **NP-40**, and **Tween-20** (any 2)

2. Name the mobile phase and one type of stationary phase used in Gas Chromatography [U] / [CO2]
Mobile phase is an inert carrier gas such as Helium or Nitrogen, Argon or Hydrogen . Stationary phase polyethylene glycol (PEG/Carbowax), Polysiloxanes (any 1 stationary and any one mobile phase)

3. Explain gel casting tray. [U] / [CO3]

A specialized mold, used in molecular biology laboratories to form agarose gels for electrophoresis. It holds liquid agarose as it cools and solidifies, forming a firm slab with sample wells, which is then transferred to an electrophoresis tank for DNA/RNA separation.

4. What is the role of fluorophore conjugated antibodies in immunophenotyping? [U] / [CO4]
Binds specifically to surface markers (antigens) on cells. When exposed to a laser, the fluorophore emits light, allowing identification and classification of different cell types.

5. List any two applications of the filtration technique. [K] / [CO1]

collection of cells, removal of precipitate, removal of particulate matter from air, clarification of juice, sterilization of media etc (any two applications of filtration technique)

6. Mention one advantage of high-performance thin-layer chromatography (HPTLC) over conventional TLC. [A] / [CO2]
HPTLC uses smaller and more uniform silica particles, which gives better separation and sharper spots compared to conventional TLC. It also requires a smaller sample volume and gives results faster (any 1)

7. What are the monomers used in preparing gel during PAGE? [U] / [CO3]

Acrylamide and N, N'-methylenebisacrylamide (bisacrylamide)

8. What is forward scatter (FSC) in flow cytometry? [K] / [CO4]

Ans: Forward Scatter (FSC) measures the amount of laser light scattered in the forward direction as a cell passes through the beam. It gives information about the size of the cell — larger cells scatter more light.

9. Explain why osmotic shock is effective mainly for Gram-negative bacteria? [K] / [CO1]

This method is highly effective against Gram-negative bacteria due to their unique, 2 layered cell wall (outer membrane and a thin peptidoglycan layer), which provides a specific location for target proteins and a "weak spot" for the shock to act on. Sudden water influx, forces the outer membrane to rupture, releasing periplasmic proteins and binding proteins.

10. List two applications of cell counting in cell culture. [U] / [CO4]
determining cell viability (live vs. dead cells) before an experiment, seeding a specific number of cells per well or flask to maintain consistent experimental conditions.

11. What is the main purpose of centrifugation in laboratories? [K] / [CO1]

Separating biological fluids, Cell Harvesting and Organelle Isolation, Separating Immiscible liquids, Molecular Purification (any 2).

12. List two types of camera lucida. [U] / [CO4]
Mirror type, which uses a mirror and prism system, and Prism type, which uses a glass prism to superimpose the image onto the drawing surface.

PART B Short Answer

Answer any 8 questions. Each question carries 5 marks.

13. Write down the principle and procedure associated with conventional filtration using filter paper. [K] / [CO1]

The principle of filtration with filter paper is based on size exclusion (mechanical sieving). The filter paper acts as a semi-permeable porous barrier. The pores in the paper are specifically designed to be smaller than the particles of the insoluble solid but larger than the solvent molecules. When a mixture is poured onto the paper, gravity forces the liquid (filtrate) through the porous structure, while the insoluble solid particles (residue) are trapped on the paper's surface or within its matrix.

PROCEDURE- Preparation and Folding, Fitting the Funnel, Wetting the Paper, setting up the apparatus, Filtration (Transferring the Solution) & Washing the Precipitate.

14. Explain the various detection methods used in chromatography. [U] / [CO2]
1. Gas Chromatography Detectors- **Flame Ionization Detector (FID):** The sample is burned in a hydrogen-air flame, producing ions. The number of ions collected is proportional to the carbon atoms in the sample. **Electron Capture Detector (ECD):** Highly sensitive to halogen-containing substances, making it ideal for detecting pesticides and insecticides. It measures the loss of current caused by analyte electron capture. **Thermal Conductivity Detector (TCD):** Detects changes in the thermal conductivity of the carrier gas due to the presence of analytes
2. High-Performance Liquid Chromatography (HPLC) Detectors- **UV-Visible Spectrophotometric Detector:** Measures the ability of analytes to absorb UV or visible light. It is widely used for compounds that contain chromophores. **Refractive Index (RI) Detector:** Measures the difference in the refractive index between the pure mobile phase and the eluent. It is a universal detector but has lower sensitivity than UV detectors. **Fluorescence Detector:** Highly sensitive and selective detector that measures the light emitted by molecules after they are excited by a specific wavelength. **Electrochemical Detector:** Measures the current resulting from the oxidation or reduction of analytes on an electrode surface.
3. UV-Visible detector — Most widely used. Detects compounds that absorb UV/visible light (e.g., proteins, nucleic acids). Simple and non-destructive. (any 5)

15. Explain the procedure of Western Blotting. [U] / [CO3]

Western blotting (immunoblotting) is a laboratory technique used to detect specific proteins in a sample, involving sample preparation, SDS-PAGE protein separation by size, transfer onto a membrane, blocking, and antibody-based detection (primary and enzyme-conjugated secondary antibodies) to visualize target proteins.

16. Explain the principle and application of micrometers in microscopy? [U] / [CO4]

Principle of Micrometry (3 marks) **-** Ocular Micrometer: A glass disc with an arbitrary, unnumbered scale (typically 100 divisions) placed inside the eyepiece. Stage Micrometer (Calibration Slide): A microscope slide with a precisely etched, known scale (usually 1 mm divided into 100 parts, where each division equals 0.01 mm or 10 µm). Calibration Principle: Because ocular divisions represent different real distances at different magnifications, the ocular micrometer must be calibrated against the stage micrometer. The relationship (ratio) between the two scales is used to calculate the actual size of the specimen.

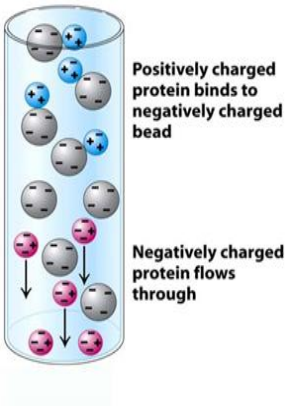
Applications in Cell Measurement (2 marks) **-** Cell Size Determination, Organelle Measurement, Microorganism Identification:

17. Write a note on refrigerated centrifuge. [K] / [CO1]

It is an essential laboratory instrument that combines high-speed centrifugation with precise temperature control (typically -20 to +40) to separate, sediment, or purify temperature-sensitive samples. It prevents heat-induced degradation of samples (e.g., DNA, RNA, proteins, enzymes) during processing. Key features include high RPM, rotor flexibility, and, for some, user-friendly interfaces.

18. With diagram explain the principle and procedure of anion exchange chromatography. [U] / [CO2]

Principle: The resin (stationary phase) is coated with positively charged groups (e.g., DEAE — diethylaminoethyl). Negatively charged molecules (anions) in a sample are attracted to and bind to these positive groups. Neutral or positively charged molecules pass through without binding. Bound anions are then released by washing with an increasing salt (e.g., NaCl) gradient or change in pH. Weakly bound anions elute first; strongly bound ones elute last.



Procedure:

Equilibrate the column with a low-salt buffer (pH depends on Isoelectric point of the molecule of interest). Load the sample onto the column.Wash with starting buffer to remove unbound molecules. Apply a salt gradient (increasing NaCl concentration)or pH gradient to elute bound anions stepwise. Collect fractions at the outlet and detect using UV absorbance or other detectors. (2 +2 +1)

19. Write a note on gel documentation system associated with electrophoresis. [K] / [CO3]

It is an essential laboratory instrument used to visualize, capture, and analyze digital images of nucleic acids or proteins separated by electrophoresis. It uses UV or blue light to excite stained gels, with a high-resolution camera recording the bands, enabling immediate quantification and data archiving. It also has white light for documenting gels stained with dyes like coomassive brilliant blue, silver staining etc

Core Components: Illumination Source, Imaging Device, Darkroom/Hood, and Analysis Software.

20. Explain the application of cell counting in hematology. [A] / [CO4]

Cell Culture and Research Applications- Cell counting techniques are fundamental to cell culture, one of the most important tools in modern biology and medicine. Accurate cell enumeration is essential for maintaining healthy cultures, conducting reproducible experiments, and ensuring consistency in biomedical research and biotechnology production.

Complete Blood Count (CBC): Counts RBCs, WBCs, and platelets in a blood sample. CBC includes red blood cell count (RBC), white blood cell count (WBC), platelet count, hemoglobin concentration, and various calculated indices. Deviations from normal ranges indicate various pathological conditions. Anemia (low RBC or hemoglobin) may result from blood loss, nutritional deficiency, or bone marrow disorders. Leukocytosis (elevated WBC) suggests infection, inflammation, or leukaemia.

21. What is differential and density gradient centrifugation? [K] / [CO1]

Differential centrifugation separates particles based on size and sedimentation rate using stepwise increases in speed, while density gradient centrifugation separates particles by density as they settle through a gradient medium. Differential Centrifugation: A homogenized sample is subjected to repeated centrifugation at increasing speeds. Larger, denser particles (e.g., nuclei) pellet first at low speeds, while smaller particles (e.g., ribosomes) remain in the supernatant and require higher speeds. Density Gradient Centrifugation: The sample is placed on top of a pre-formed gradient (such as sucrose) or in self-forming fgradient like caesium chloride. Particles migrate to positions where their density matches that of the surrounding medium.

22. Explain the concept and applications of single cell analysis. [U] / [CO4]

Single-cell analysis examines the genomic, transcriptomic, or proteomic profiles of individual cells, revealing cell-to-cell heterogeneity within populations that is masked by traditional bulk methods. It identifies rare cell types, and maps cellular trajectories in development and disease (1 marks)

Applications: (4 marks)

Cancer research: Identifies rare drug-resistant tumor cells hidden within a tumor mass. Helps understand tumor evolution and heterogeneity.

Developmental biology: Tracks gene expression changes as a single stem cell differentiates into specialized cell types (e.g., blood cells, neurons).

Immunology: Profiles individual immune cells (T cells, B cells, NK cells) during infection or vaccination to understand immune responses.

Neuroscience: Studies individual neuron types and their specific gene expression patterns in the brain.

Microbiology: Analyzes individual bacteria in a mixed community (microbiome) to identify species that cannot be cultured.

Clinical diagnostics: Identifies circulating tumor cells (CTCs) in blood — a liquid biopsy for cancer diagnosis.

PART C Answer any 1 questions

Each question carries 10 marks.

23. Discussion in detail the principle, instrumentation and working of HPLC. Compare normal phase and reverse phase HPLC. [U] / [CO2]

Principle (2 marks)

HPLC is an advanced chromatographic technique that separates compounds based on their differential partitioning between a liquid mobile phase and a solid stationary phase packed in a column.

The fundamental principle involves passing a liquid solvent containing the sample mixture through a column filled with very fine particles under high pressure. Different components in the mixture interact differently with the stationary phase based on their chemical properties such as polarity, size, or charge.

Components that interact weakly with the stationary phase move through the column faster and elute earlier, while those with stronger interactions are retained longer and elute later. This difference in retention

times allows the mixture to be separated into individual components. The separation can occur through various mechanisms including adsorption, partition, ion exchange, or size exclusion, depending on the type of column and conditions used.

Instrumentation and Working (4 marks)

The HPLC system consists of several key components that work together to achieve separation. A solvent reservoir holds the mobile phase, which can be a single solvent or a mixture of solvents. The pump is a critical component that delivers the mobile phase at high pressure (typically 1000-6000 psi) with precise and constant flow rates. This high pressure is necessary to push the solvent through the tightly packed column containing very fine particles. The sample is introduced into the system through an injector, which allows precise volumes to be loaded onto the column without interrupting the solvent flow. The heart of the system is the column, which is usually a stainless-steel tube packed with the stationary phase material. As the mobile phase carries the sample through the column, different components separate based on their retention times. After leaving the column, the separated components pass through a detector that identifies and measures them. Common detectors include UV-visible detectors that measure light absorption, fluorescence detectors for fluorescent compounds, and refractive index detectors for compounds without chromophores. The detector sends signals to a computer system that records and displays the data as a chromatogram, showing peaks for each separated component. The chromatogram displays detector response versus time, where each peak represents a different compound, the peak height or area indicates the amount present, and the retention time helps identify the compound.

Comparison Table: Normal Phase vs. Reverse Phase HPLC (4 marks)

Parameter	Normal Phase HPLC	Reverse Phase HPLC
Stationary Phase	Polar (e.g., Silica)	Non-polar (e.g., C18, C8)
Mobile Phase	Non-polar (e.g., Hexane)	Polar (e.g., Water, Methanol)
Compounds Retained	Polar compounds	Non-polar compounds
Elution Order	Non-polar elutes first	Polar elutes first
Sample Type	Non-aqueous, polar samples	Aqueous, non-polar samples
Sensitivity to Water	High (water can damage column)	Low (water is the mobile phase)
Column Equilibration	Slower	Faster
Usage Frequency	Less common (~20-30%)	Very common (~70-80%)
Applications	Lipids, fat-soluble vitamins, isomers	Drugs, amino acids, water-soluble vitamins

24. Bring out the applications, advantages, and limitations of 2D Electrophoresis? [U] / [CO3]

Applications (6 marks)

Proteomics and Protein Expression Profiling (compares protein expression levels between samples (e.g., diseased vs. healthy cells, treated vs. untreated samples) to identify upregulated or downregulated proteins), Biomarker Discovery (to identify specific protein markers in serum, tissue, or bodily fluids for cancer detection and other diseases), Post-Translational Modification (PTM) Analysis (protein modifications like phosphorylation, glycosylation, methylation, and acetylation), Protein Interaction Studies (used to study interactions by separating complexes and observing shifts in protein spots), Proteome Mapping and Characterization: (map the entire proteome of organelles, bacteria, or cells, allowing for the cataloging of protein responses to environmental changes) Food Safety and Adulteration (verify product purity in pharmaceutical manufacturing, analyze batch-to-batch consistency, and detect food adulteration), and Microbial Identification (based on unique proteins)

Advantages (2 marks)

High Resolution, Quantitative Analysis, Low Cost, and Visual Map of Intact Proteins

Limitations (2 marks)

Technical Complexity and Time, Poor Reproducibility, and Limited Detection of Specific Proteins